

Exp: 02

Write the Python Program to solve 8-Queen Problem

Aim:

To write a python program to solve 8-Queen problem.

OBJECTIVE:

To understand the backtracking search technique by placing 8 queens on an 8x8 chessboard such that no two queens attack each other.

Algorithm:

Step1: Start with an empty board of size 8x8.

Step2: Place the first queen in the first row and the first column.

Step3: For each subsequent row, check all possible columns to place the queen, and backtrack if a valid position cannot be found.

Step4: A valid position is a column where no other queens are in the same row, column, or diagonal.

Step5: If no solution is found, return failure

Program:

N = 8

```
board = [[0] * N for _ in range(N)]
```

```
def is_safe(row, col):
```

```
    for i in range(col):
```

```
        if board[row][i]:
```

```
            return False
```

```
    i, j = row, col
```

```
    while i >= 0 and j >= 0:
```

```
        if board[i][j]:
```

```
            return False
```

```
        i -= 1
```

```
        j -= 1
```

```

i, j = row, col
while i < N and j >= 0:
    if board[i][j]:
        return False
    i += 1
    j -= 1

return True

def solve(col):
    if col >= N:
        return True

    for row in range(N):
        if is_safe(row, col):
            board[row][col] = 1

            if solve(col + 1):
                return True

            board[row][col] = 0

    return False

if solve(0):
    print("Solution:")
    for row in board:
        print(*row)
else:
    print("No solution exists")

```

Output:

```
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0
0 0 0 0 0 1 0 0
0 0 1 0 0 0 0 0
```

Result: Hence, the simple python program for 8- Queen problem is done